X-CAVATE Protocol

G-Code Generation from Vascular Network Geometry

Stage II of the Vascular Network 3D Bioprinting Workflow

This protocol describes how to use X-CAVATE to convert SimVascular b-spline output into collision-free 3D printer G-code. X-CAVATE can be run in three modes: a **web GUI** (recommended for new users), a **command-line interface** (CLI), or via **Docker**. The steps below cover all three.

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1 Part A: Installation (5 minutes)

1.1 Option 1: Conda (Recommended)

1. Open a terminal (Mac: Terminal.app; Windows: Anaconda Prompt or PowerShell).
2. Confirm conda is installed:

   ```
   conda --version
   ```

   If not installed, download Anaconda or Miniconda from https://docs.conda.io/en/latest/miniconda.html.
3. Navigate to the folder where you want to store X-CAVATE and clone the repository:

   ```
   git clone https://github.com/skylarscottlab/X-CAVATE.git
   cd X-CAVATE
   ```

   Alternatively, download and unzip the repository from the GitHub page (green "Code" button → "Download ZIP").
4. Create and activate the conda environment (installs Python 3.11, all dependencies, and the X-CAVATE package):

   ```
   conda env create -f environment.yml
   conda activate xcavate-gui
   ```
5. Verify the installation:

   ```
   xcavate --help
   ```

   You should see the full list of command-line parameters.

To remove the environment later: `conda env remove -n xcavate-gui`.

1.2 Option 2: pip install

1. Open a terminal (Mac: Terminal.app; Windows: Anaconda Prompt or PowerShell).
2. Confirm Python 3.9+ is installed:

   ```
   python3 --version
   ```
3. Clone and enter the repository:

   ```
   git clone https://github.com/skylarscottlab/X-CAVATE.git
   cd X-CAVATE
   ```

4. Install X-CAVATE with GUI support (includes numpy, scipy, pandas, plotly, matplotlib, streamlit):

   ```
   pip install ".[gui]"
   ```

   If you only need the CLI (no web interface):

   ```
   pip install .
   ```

5. Verify the installation:

   ```
   xcavate --help
   ```

1.3 Option 3: Docker (no Python required)

1. Install Docker Desktop from https://www.docker.com/products/docker-desktop.
2. Open a terminal and navigate to the X-CAVATE folder:

   ```
   cd X-CAVATE
   ```

3. Build the Docker image:

   ```
   docker build -t xcavate .
   ```

4. Run the container:

   ```
   docker run -p 8501:8501 xcavate
   ```

5. Open your browser to **http://localhost:8501**. The X-CAVATE web interface should appear.

2 Part B: Prepare Input Files (5 minutes)

X-CAVATE requires two input files from your SimVascular workflow.

2.1 B.1 Network File (b-splines)

This file contains the vessel centerline coordinates exported from SimVascular. Each vessel has a header line followed by coordinate data rows.

```
Vessel: 0, Number of Points: 100
33.4878, 0.0000, 53.5303, 0.0200
33.4115, -0.5285, 52.7311, 0.0200
...
```

```
Vessel: 1, Number of Points: 100
15.0600, -37.3300, 11.6600, 0.0190
...
```

- **Columns 1–3** (required): x, y, z coordinates in centimeters.
- **Column 4** (optional): vessel radius, needed if using speed calculation.
- **Column 5** (optional): arterial/venous flag (0 = venous, nonzero = arterial), needed for multimaterial printing.

CAUTION: Coordinates must be in centimeters. X-CAVATE internally converts to millimeters.

CAUTION: Limit each vessel to no more than 100 coordinates to avoid long runtimes.

2.2 B.2 Inlet/Outlet File

This file identifies which network locations are inlets (where printing begins) and outlets.

```
inlet
21.97, -15.43, 50.00
11.81, 0.00, 21.34
outlet
36.05, -27.19, 50.00
50.00, 0.00, 26.54
```

Each section begins with the keyword `inlet` or `outlet` on its own line, followed by one or more (x, y, z) coordinate triplets.

Note: Inlet/outlet coordinates are matched to the nearest network point automatically using a KD-tree nearest-neighbor search. They do not need to exactly match a coordinate in the network file. A warning is displayed if the nearest match is more than 5.0 mm away.

2.3 B.3 File Placement

- **GUI or Docker mode:** Files are uploaded through the web interface. No file placement needed.
- **CLI mode:** Place both files in the `inputs/` folder inside the X-CAVATE directory (or specify full paths via `-network_file` and `-inletoutlet_file`).

3 Part C: Custom G-Code Templates (5 minutes, optional)

Custom G-code templates adapt X-CAVATE's output to your specific printer hardware and software. Each template file contains a G-code snippet that is automatically inserted at the appropriate location in the final G-code output.

The `inputs/custom/` folder contains 12 template files with placeholder text. Replace the placeholder text with your own G-code for each section relevant to your printer.

3.1 C.1 Header

File	Instructions
`header_code.txt`	Replace the placeholder text with your custom G-code for the header section of the final G-code output. Include any code required for the functioning of your printer, such as establishing connections between external pressure boxes and syringes.

3.2 C.2 Single Material Extrusion

Skip this section if printing multimaterial; proceed to C.3.

File	Instructions
`start_extrusion_code.txt`	Replace with your G-code for starting extrusion.
`stop_extrusion_code.txt`	Replace with your G-code for stopping extrusion.

3.3 C.3 Multimaterial Extrusion

Skip this section if printing single material; proceed to C.4.

File	Instructions
`start_extrusion_code_printhead1.txt`	Start extrusion for Printhead 1 (arterial).
`start_extrusion_code_printhead2.txt`	Start extrusion for Printhead 2 (venous).
`stop_extrusion_code_printhead1.txt`	Stop extrusion for Printhead 1 (arterial).
`stop_extrusion_code_printhead2.txt`	Stop extrusion for Printhead 2 (venous).
`active_pressure_printhead1.txt`	Set ink pressure when PH1 is the **active** printhead.
`active_pressure_printhead2.txt`	Set ink pressure when PH2 is the **active** printhead.
`rest_pressure_printhead1.txt`	Set ink pressure when PH1 is **inactive**.
`rest_pressure_printhead2.txt`	Set ink pressure when PH2 is **inactive**.

3.4 C.4 Dwell (Optional)

Incorporating time delays ("dwells") at the first and last node of each print pass deposits additional ink at each junction, improving the connection via a process similar to "spot welding."

File	Instructions
`dwell_code.txt`	Replace with your G-code for creating a time delay at the start and end of each print pass (e.g., `G4 P80` for an 80 ms dwell).

3.5 C.5 How to Provide Custom G-Code in Each Mode

Mode	Instructions
GUI	Enable the **Custom G-code** toggle in the sidebar's Advanced section. A tabbed editor (Header, Single Material, Multimaterial, Dwell) appears in the main area. Paste your G-code into the text areas.
CLI	Edit the `.txt` files in `inputs/custom/` directly with a text editor, then run with `-custom 1`.

Docker	*Option A:* Use the GUI editor in the browser. *Option B:* Mount your local `inputs/custom/` folder: `docker run -p 8501:8501 -v $(pwd)/inputs/custom:/app/inputs/custom xcavate`, then enable **Custom G-code** in the GUI.

4 Part D: Execute X-CAVATE

4.1 Mode 1: Web GUI

1. Activate the environment and start the server:

```
conda activate xcavate-gui
streamlit run xcavate/gui/app.py
```

   A browser window opens at **http://localhost:8501**.

2. In the **sidebar**, upload your two input files and configure all parameters (see Section 4.1.1 for a complete reference).
3. Click the Run X-CAVATE button (blue, full-width, at the bottom of the sidebar). This button is disabled until both input files are uploaded.
4. A progress bar in the main area tracks the pipeline through 7 steps:
   (a) Reading network and inlet/outlet files
   (b) Interpolating network to nozzle resolution
   (c) Building adjacency graph and detecting branchpoints
   (d) Generating collision-free print passes (DFS)
   (e) Post-processing: subdivision, gap closure, overlap
   (f) Processing multimaterial passes (if enabled)
   (g) Writing output files and G-code
5. When complete, the main area displays:
   - **3D Visualization** — Interactive Plotly scatter plots of the single-material and (if enabled) multimaterial print passes. Rotate, zoom, and inspect paths directly in the browser.
   - **Download buttons** — Organized in three columns:
     - **G-code** column: one download button per G-code file (e.g., `gcode_SM_pressure.txt`).
     - **Coordinate Files** column: download buttons for all coordinate files (`x_coordinates_SM.txt`, etc.), speed files, and radii files.
     - **Changelog** column: download button for `changelog.txt`.
6. After the run completes, **Print Instructions** appear in the terminal console showing nozzle positioning, G92 zeroing coordinates, and container centering information.

4.1.1 GUI Sidebar Reference

The sidebar is organized into the following sections. Every widget, its type, default value, and purpose are listed below.

Input Files

Widget	Type	Description
Network file (.txt)	File uploader	Upload the network coordinates file (see Part B.1). Accepts `.txt` files only.
Inlet / Outlet file (.txt)	File uploader	Upload the inlet/outlet file (see Part B.2). Accepts `.txt` files only.

Required Parameters

Widget	Type	Default	Description
Nozzle diameter (mm)	Number	0.500	Outer diameter of the print nozzle. Controls interpolation density (nozzle radius = half this value).
Container height (mm)	Number	10.0	Height of the print container. Used for nozzle raise/lower calculations.
Decimal places	Integer	3	Number of decimal places for output coordinate rounding (0–10).
Amount up (mm)	Number	10.0	Distance to raise nozzle above the container between print passes.
Printer type	Selectbox	Pressure	Dropdown with three options: **Pressure** (syringe-based), **Positive Ink** (displacement), **Aerotech** (6-axis controller).
Multimaterial	Toggle	Off	Enable dual-ink printing. Requires column 5 (artven flag) in the network file.

Optional Parameters — Tolerancing (expandable)

Widget	Type	Default	Description
Enable tolerance	Toggle	Off	Adds a tolerance offset to vessel dimensions.
Tolerance	Number	0.00	Tolerance amount in mm. Only visible when toggle is on.

Optional Parameters — Print Settings (expandable)

Widget	Type	Default	Description
Print speed (mm/s)	Number	1.000	Default print speed for constant-speed mode.
Flow (mm^3/s)	Number	0.161	Volumetric flow rate for speed calculation. Used to compute per-node speeds: $v = flow/(\pi r^2)$.
Jog speed (mm/s)	Number	5.00	Speed for XY travel moves between passes.
Jog speed lift (mm/s)	Number	0.25	Speed for initial Z-lift at end of a pass.
Initial lift (mm)	Number	0.50	Distance for the initial lift before jogging to network top.
Dwell start (s)	Number	0.080	Pause duration at the start of each pass ("spot welding").
Dwell end (s)	Number	0.080	Pause duration at the end of each pass.

Optional Parameters — Geometry (expandable)

Widget	Type	Default	Description
Container X (mm)	Number	50.0	Container x-dimension. Used in print instructions for centering.
Container Y (mm)	Number	50.0	Container y-dimension. Used in print instructions for centering.
Scale factor	Number	1.000	Multiplier applied to all network coordinates after import.
Top padding (mm)	Number	0.0	Extra padding above the highest network point.

Optional Parameters — Downsampling (expandable)

Widget	Type	Default	Description
Enable downsampling	Toggle	Off	Reduces coordinate density after processing by keeping every Nth node while preserving branchpoints/endpoints.

Downsample factor	Integer	1	Keep every Nth node. Only visible when toggle is on.

Optional Parameters — Multimaterial (expandable) These settings only apply when the **Multimaterial** toggle is on.

Widget	Type	Default	Description
Offset X (mm)	Number	103.0	X-distance between printhead nozzle tips.
Offset Y (mm)	Number	0.5	Y-distance between printhead nozzle tips.
Front nozzle	Integer	1	Which nozzle is in front: 1 = venous in front, 2 = venous behind.
Printhead 1 name	Text	Aa	Name for the arterial printhead (used in G-code).
Printhead 2 name	Text	Ab	Name for the venous printhead.
Axis 1 name	Text	A	Z-axis name for the arterial printhead.
Axis 2 name	Text	B	Z-axis name for the venous printhead.
Resting pressure (psi)	Number	0.0	Pressure when a printhead is inactive.
Active pressure (psi)	Number	5.0	Pressure when a printhead is actively extruding.

Optional Parameters — Positive Ink Displacement (expandable) These settings only apply when **Printer type** is set to **Positive Ink**.

Widget	Type	Default	Description
Start	Number	0.00	Syringe displacement at pass start (SM).
End	Number	0.00	Syringe displacement at pass end (SM).
Use radii	Toggle	Off	Compute displacement from vessel radii.
Diameter	Number	1.000	Nozzle inner diameter for displacement calc.
Syringe diameter	Number	1.000	Syringe barrel diameter.
Factor	Number	1.000	Multiplier on computed displacement.
Start (arterial)	Number	0.00	Pass start displacement for arterial (MM).

Start (venous)	Number	0.00	Pass start displacement for venous (MM).
End (arterial)	Number	0.00	Pass end displacement for arterial (MM).
End (venous)	Number	0.00	Pass end displacement for venous (MM).

Optional Parameters — Advanced (expandable)

Widget	Type	Default	Description
Custom G-code	Toggle	Off	Enable custom G-code templates. When on, a tabbed editor appears in the main area with four tabs: **Header**, **Single Material**, **Multimaterial**, **Dwell**. Paste your G-code snippets into the text areas (see Part C).
Pathfinding algorithm	Selectbox	DFS	**DFS**: modified depth-first search with KD-tree collision detection.
Overlap nodes	Integer	0	Number of overlap nodes to add at pass junctions (0 = disabled). Improves print continuity at branchpoints.
Overlap algorithm	Selectbox	Retrace	Only visible when overlap > 0. **Retrace**: scans all previous passes for shared nodes, retraces backwards, appends to end of current pass.
Generate plots	Toggle	On	Generate interactive 3D Plotly visualizations.
Speed calculation	Toggle	Off	Compute per-node print speeds based on vessel radii and flow rate: $v = flow/(\pi r^2)$. Requires column 4 (radius) in the network file.

4.1.2 Custom G-Code Template Editor (GUI)

When the **Custom G-code** toggle is on, the main area shows a tabbed editor with four tabs:

1. **Header tab** — A single text area labeled **Header code**. Paste G-code that will appear at the top of the output file (e.g., homing, fan control, connection setup).
2. **Single Material tab** — Two text areas side by side:

- **Start extrusion** (left): G-code to begin extrusion at each pass start.
- **Stop extrusion** (right): G-code to stop extrusion at each pass end.

3. **Multimaterial tab** — Eight text areas in two sections:
   - **Extrusion Start / Stop**: four text areas for start/stop extrusion on Printhead 1 and Printhead 2.
   - **Pressure Control**: four text areas for active/resting pressure on Printhead 1 and Printhead 2. Each has a help tooltip explaining when it applies.
4. **Dwell tab** — A single text area labeled **Dwell code (applied at start and end of each pass)**. Example: `G4 P80` for an 80 ms pause.

4.2 Mode 2: Command-Line Interface

1. Open a terminal, activate the environment, and navigate to the X-CAVATE directory:

```
conda activate xcavate-gui
cd X-CAVATE
```

2. (If using custom G-code) Edit the template files in `inputs/custom/` (see Part C), then include `-custom 1` in the command.
3. Run X-CAVATE with your parameters. Example for a single-material pressure print:

```
xcavate \
  --network_file inputs/VesselNetwork.txt \
  --inletoutlet_file inputs/InletsOutlets.txt \
  --nozzle_diameter 0.65 \
  --container_height 50.0 \
  --num_decimals 5 \
  --amount_up 10.0 \
  --printer_type 0 \
  --multimaterial 0 \
  --tolerance_flag 0 \
  --speed_calc 1 \
  --plots 1 \
  --downsample 0 \
  --custom 1 \
  --flow 0.161
```

   Or equivalently: `python -m xcavate [same arguments]`
4. The terminal outputs progress through each pipeline step, followed by **Print Instructions** with nozzle positioning, G92 zeroing, container centering, and padding values.
5. Output files are written to `outputs/` (or the directory specified by `-output_dir`).

4.2.1 Complete CLI Parameter Reference

Required Parameters

Parameter	Type	Example	Description
`-network_file`	str	`inputs/net.txt`	Path to network coordinates file.
`-inletoutlet_file`	str	`inputs/io.txt`	Path to inlet/outlet file.
`-nozzle_diameter`	float	`0.65`	Nozzle outer diameter (mm).
`-container_height`	float	`50`	Container height (mm).
`-num_decimals`	int	`5`	Decimal places for rounding.
`-amount_up`	float	`10`	Z-raise between passes (mm).
`-printer_type`	int	`0`	0 = Pressure, 1 = Positive Ink, 2 = Aerotech.
`-multimaterial`	int	`0`	0 = single ink, 1 = dual ink.
`-tolerance_flag`	int	`0`	0 = off, 1 = on.
`-speed_calc`	int	`1`	0 = constant speed, 1 = radius-based speed.
`-plots`	int	`1`	0 = no plots, 1 = generate HTML plots.
`-downsample`	int	`0`	0 = off, 1 = on.
`-custom`	int	`1`	0 = no custom G-code, 1 = use templates.

Optional — Geometry

Parameter	Type	Default	Description
`-container_x`	float	50	Container x-dimension (mm).
`-container_y`	float	50	Container y-dimension (mm).
`-scale_factor`	float	1.0	Multiplier on all coordinates.
`-top_padding`	float	0	Extra padding above network (mm).
`-tolerance`	float	0	Tolerance amount (mm).

Optional — Print Settings

Parameter	Type	Default	Description
`-print_speed`	float	1.0	Constant print speed (mm/s).
`-flow`	float	0.161	Volumetric flow rate (mm^3/s). Experimentally determined default.
`-jog_speed`	float	5.0	XY travel speed (mm/s).
`-jog_speed_lift`	float	0.25	Z-lift speed (mm/s).
`-initial_lift`	float	0.5	Initial lift distance (mm).
`-jog_translation`	float	10	Speed between nozzles for MM (mm/s).
`-dwell_start`	float	0.08	Dwell at pass start (seconds).

`-dwell_end`	float	0.08	Dwell at pass end (seconds).

Optional — Multimaterial

Parameter	Type	Default	Description
`-offset_x`	float	103	Printhead X-offset (mm).
`-offset_y`	float	0.5	Printhead Y-offset (mm).
`-front_nozzle`	int	1	1 = venous in front, 2 = behind.
`-printhead_1`	str	Aa	Arterial printhead name.
`-printhead_2`	str	Ab	Venous printhead name.
`-axis_1`	str	A	Arterial Z-axis name in G-code.
`-axis_2`	str	B	Venous Z-axis name in G-code.
`-resting_pressure`	float	0	Inactive printhead pressure (psi).
`-active_pressure`	float	5	Active printhead pressure (psi).

Optional — Pathfinding and Post-processing

Parameter	Type	Default	Description
`-algorithm`	str	dfs	Pathfinding: `dfs` or `sweep_line`.
`-num_overlap`	int	0	Number of overlap nodes at junctions.
`-overlap_algorithm`	str	retrace	`retrace`: scans all previous passes (original).
`-downsample_factor`	int	1	Keep every Nth node.
`-close_sm`	int	0	Use SM gap closure file (0/1).
`-close_mm`	int	0	Use MM gap closure file (0/1).
`-output_dir`	str	outputs	Output directory path.

Optional — Positive Ink Displacement

Parameter	Type	Default	Description
`-positiveInk_start`	float	0	Syringe displacement at pass start (SM).
`-positiveInk_end`	float	0	Syringe displacement at pass end (SM).
`-positiveInk_radii`	int	0	Use radii for displacement (0/1).
`-positiveInk_diam`	float	1	Nozzle inner diameter (mm).

`-positiveInk_syringe_diam`	float	1	Syringe barrel diameter (mm).
`-positiveInk_factor`	float	1	Displacement multiplier.
`-positiveInk_start_arterial`	float	0	Pass start displacement, arterial (MM).
`-positiveInk_start_venous`	float	0	Pass start displacement, venous (MM).
`-positiveInk_end_arterial`	float	0	Pass end displacement, arterial (MM).
`-positiveInk_end_venous`	float	0	Pass end displacement, venous (MM).

4.3 Mode 3: Docker

1. Build and start the container (if not already running):

```
docker build -t xcavate .
docker run -p 8501:8501 -v $(pwd)/outputs:/app/outputs xcavate
```

2. Open **http://localhost:8501** in your browser.
3. Follow the same steps as **Mode 1 (Web GUI)** above. All functionality is identical.
4. To use pre-written custom G-code template files, mount the `inputs/custom/` folder:

```
docker run -p 8501:8501 \
  -v $(pwd)/inputs/custom:/app/inputs/custom \
  -v $(pwd)/outputs:/app/outputs \
  xcavate
```

5. Output files are accessible in the mounted `outputs/` directory or via the download buttons.

5 Part E: Inspect Output Files

After execution completes, X-CAVATE produces the following outputs.

5.1 E.1 G-Code Files

Located in `outputs/gcode/`:

File	Description
`gcode_SM_pressure.txt`	Single-material G-code for Pressure printer.
`gcode_SM_positive_ink.txt`	Single-material G-code for Positive Ink printer.
`gcode_SM_aerotech.txt`	Single-material G-code for Aerotech printer.
`gcode_MM_*.txt`	Multimaterial equivalents (generated if multimaterial is enabled).

The filename suffix matches your selected printer type.

5.2 E.2 Coordinate Files

Located in `outputs/graph/`:

File	Description
`x_coordinates_SM.txt`	X-coordinates, one line per node, separated by pass.
`y_coordinates_SM.txt`	Y-coordinates per pass.
`z_coordinates_SM.txt`	Z-coordinates per pass.
`all_coordinates_SM.txt`	Combined x, y, z per pass.
`radii_list_SM.txt`	Vessel radii per node per pass.
`printspeed_list_SM.txt`	Computed print speeds per node (if speed_calc enabled).
`*_MM.txt`	Multimaterial equivalents (if enabled).
`graph.txt`	Full adjacency graph.
`special_nodes.txt`	Branchpoints, endpoints, inlets, outlets with coordinates.

5.3 E.3 Visualizations

Located in `outputs/plots/` (when plots are enabled):

File	Description
`network_original.html`	Pre-interpolation network, one trace per vessel.
`network_SM.html`	Single-material print passes (post-processing).
`network_downsampled_SM.html`	Downsampled SM passes (if downsampling enabled).
`network_SM_overlap.html`	SM passes with overlap nodes (if overlap > 0).
`network_MM.html`	Multimaterial passes colored by arterial/venous (if enabled).

Open the `.html` files in any web browser to rotate, zoom, and inspect the print paths.

5.4 E.4 Changelog

The file `outputs/changelog.txt` contains a log of all gap closure operations performed during execution.

5.5 E.5 Print Instructions (Console Output)

After G-code generation, X-CAVATE prints operator instructions to the terminal (CLI) or console (GUI). These include:

- **Single material:** Starting nozzle position (`X_start`, `Y_start`), G92 zeroing command, G1 command to move to the first print coordinate, Z positioning instructions.
- **Multimaterial calibration:** Step-by-step procedure for measuring X/Y offsets between two nozzle tips using a calibration point.
- **Multimaterial printing:** Dual-nozzle positioning, dual-axis G92 zeroing, Z-alignment procedure.
- **Padding report:** Left, right, back, and front distances from the network bounding box to the container edges.

6 Troubleshooting

Problem	Solution
`Runtime instance already exists` (GUI)	A stale Streamlit process is running. Kill it: `pkill -f streamlit`, then restart.
Inlet/outlet matching warnings	The inlet/outlet coordinates are approximate. X-CAVATE matches to the nearest network point via KD-tree and logs the distance. Verify the matches are reasonable (warning appears if distance $> 5\,\mathrm{mm}$).
Pipeline hangs on large networks	Reduce the number of coordinates per vessel to 100 or fewer, or enable downsampling.
`ModuleNotFoundError`	Run `pip install ".[gui]"` to install all dependencies.
Docker container won't start	Ensure Docker Desktop is running and port 8501 is not in use.

7 Quick Reference: Mode Comparison

Feature	GUI	CLI	Docker
Python required	Yes (conda or pip)	Yes (conda or pip)	No
Custom G-code via files	No (use GUI editor)	Yes (`inputs/custom/`)	Optional (mount or GUI)
Custom G-code via editor	Yes (tabbed editor)	No	Yes (tabbed editor)
Interactive 3D plots	In-browser (after run)	Open `.html` files	In-browser (after run)
File download	Download buttons	Files in `outputs/`	Buttons + mounted volume
Print instructions	Console output	Terminal output	Console output
Overlap algorithm	Sidebar selectbox	`-overlap_algorithm`	Sidebar selectbox
Pathfinding selection	Sidebar selectbox	`-algorithm`	Sidebar selectbox
Best for	New users, exploration	Scripting, batch jobs	Reproducibility, sharing